Does Closing the Modality Gap Help Hard-Negative-Trained CLIP?

Abstract

Closing the modality gap does not improve clean compositional accuracy of hard-negative-trained CLIP; it buys a small gain in robustness to embedding noise at the cost of order sensitivity. CLIP places image and text embeddings in two separate regions of its shared space (the modality gap). NegCLIP fine-tunes CLIP with hard negative captions and images to make it compositional. We measured the gap in OpenAI CLIP ViT-B/32, the released NegCLIP and four models we fine-tuned ourselves in a 2×2 grid (with or without hard negative images × with or without hard negative captions). We then shifted embeddings post hoc to shrink the gap and evaluated each setting on ARO (VG-Relation, VG-Attribution, COCO-Order) and COCO retrieval, clean and under Gaussian embedding noise.

Three findings stand out. First, compositionality comes from hard negative captions, while the size of the gap is driven by hard negative images and the learning rate, so the two are largely decoupled. Second, closing half the gap lowers COCO-Order accuracy by 0.07–0.14 for every order-sensitive model, because re-normalising shifted text embeddings favours word-shuffled captions, which sit further along the gap direction. Third, under noise a partially closed gap (about 0.4–0.6 remaining) gives the best retrieval for every model, with a slightly larger gain for hard-negative models. Our one-file reproduction of NegCLIP matches the released checkpoint on every benchmark.

1. Introduction to CLIP

1.1 Contrastive representation learning

Contrastive learning trains an encoder without labels by pulling together representations of two views of the same input and pushing apart views of different inputs. The standard objective is InfoNCE (Oord et al., 2018), used by SimCLR and MoCo for images. For an anchor u with its positive v^+ and a set of negatives, with cosine similarity s and temperature τ:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(s(u, v^{+})/\tau)}{\sum_{v \in \{v^{+}\} \cup \mathcal{N}} \exp(s(u, v)/\tau)}$$

Wang and Isola (2020) showed that this loss decomposes into two properties on the unit hypersphere: alignment (positive pairs close together) and uniformity (all embeddings spread evenly). Both appear as diagnostics in this report. How well the representation transfers depends on the augmentations that define the views, the batch size (the number of in-batch negatives) and the temperature.

1.2 CLIP

CLIP (Radford et al., 2021) applies the same idea across two modalities. An image encoder f (here a ViT-B/32) and a text encoder g (a 12-layer causal Transformer) each map their input to a 512-dimensional vector, which is L2-normalised. For a batch of N image–caption pairs, CLIP computes the N × N matrix of cosine similarities, scales it by a learned logit scale ($1/\tau$, capped at 100), and applies cross-entropy in both directions. Each image must pick its own caption among the N captions, and each caption its own image:

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2N} \sum_{i=1}^{N} \left[-\log \frac{e^{x_i^\top y_i/\tau}}{\sum_{j} e^{x_i^\top y_j/\tau}} - \log \frac{e^{x_i^\top y_i/\tau}}{\sum_{j} e^{x_j^\top y_i/\tau}} \right]$$

Trained on 400M web image–text pairs, CLIP supports zero-shot classification (compare an image with prompts such as "a photo of a dog") and image–text retrieval without task-specific training.

1.3 The modality gap

Liang et al. (2022) found that CLIP's image and text embeddings occupy two separate, narrow cones of the sphere. They measure the gap as the distance between the two centroids:

$$\Delta = \frac{1}{n}\sum_{i=1}^{n} x_i - \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \text{gap} = \lVert \Delta \rVert_2$$

For CLIP ViT-B/32 on 5,000 COCO pairs, $\|\Delta\| = 0.82$. The gap starts at random initialisation (each encoder maps inputs into its own cone) and is preserved by the contrastive loss at low temperature. Shifting embeddings along Δ changes zero-shot accuracy and fairness. Chowers et al. (2026) report that the gap can behave like a global vector nearly orthogonal to the embeddings, and that closing it post hoc improves robustness to embedding perturbations without changing clean accuracy.

1.4 Bag-of-words behaviour and NegCLIP

Yuksekgonul et al. (2023) showed that CLIP behaves like a bag of words: it barely distinguishes "the horse is eating the grass" from "the grass is eating the horse". Their ARO benchmark measures this, and their fix, NegCLIP, fine-tunes CLIP on COCO with composition-aware hard negatives:

- **Hard negative captions:** each caption gets word-swapped versions (nouns, adjectives, verb phrases swapped). They are added as extra columns of the similarity matrix, so they act only as image→text negatives.
- **Hard negative images:** each image's 3 nearest-neighbour COCO images (by CLIP similarity) are candidates; one is sampled into the batch with its own caption.

NegCLIP raises VG-Relation from 0.63 to 0.81 and COCO-Order from 0.46 to 0.86. Kamath et al. (2024) later showed that such models can over-react to harmless rewordings ("hard positives").

2. Problem statement

Research question: does closing the modality gap help hard-negative-trained CLIP, and does hard-negative training itself change the gap?

Two lines of work meet here without having been connected. Closing the gap is reported to make CLIP more robust without hurting accuracy (Chowers et al., 2026). Hard-negative fine-tuning makes CLIP compositional (Yuksekgonul et al., 2023). Nobody has asked how the two interact. Hard negative captions are, by construction, captions that lie close to the true caption in text space, so the geometry of the text side, and with it the gap, could change. We break the question into four parts:

1. **Measure:** how large is the gap in CLIP and NegCLIP, and how is it shaped?
2. **Alter:** what happens to compositional accuracy, retrieval and noise robustness when the gap is shrunk or widened post hoc?
3. **Control:** which ingredient of NegCLIP (COCO fine-tuning, hard negative images, hard negative captions) is responsible for any difference?
4. **Explain:** why does a shift change (or not change) which caption wins on each benchmark?

Hypotheses going in:

- H1: hard-negative training changes the gap, because it reshapes the text embeddings.
- H2: following Chowers et al., closing the gap improves robustness to noise without hurting clean accuracy, for NegCLIP as for CLIP.

- H3: because image→text benchmarks compare captions for the same image, a uniform shift of text embeddings barely changes which caption wins.

The mid-project targets from the proposal were: (T1) gap measured for OpenCLIP and NegCLIP; (T2) a gap shift implemented with first results on OpenCLIP; (T3) an evaluation pipeline on a compositionality benchmark. All three are done, and the fine-tuned controls planned for the final presentation are also done.

3. Datasets

All data comes from MS-COCO 2014 and Visual Genome, split so that nothing we evaluate on was ever trained on.

Dataset	Size	Used for	Source
COCO Karpathy val	5,000 images, 25,010 captions	Fitting the gap vector Δ (held out); NegCLIP's checkpoint selection	COCO val2014 + BLIP Karpathy json
COCO Karpathy test	5,000 images, 25,010 captions	Gap measurement; image↔text retrieval (Recall@1/5/10)	COCO val2014 + BLIP Karpathy json
NegCLIP train split	109,468 images, 1 caption each (80,041 train2014 + 29,427 val2014 "restval")	Fine-tuning all four controls	ARO repo, train_neg_clip.tsv
NegCLIP valid split	4,839 images (= Karpathy val)	LR selection, per-epoch gap tracking	ARO repo, valid_neg_clip.tsv
ARO VG-Relation	23,937 cases (13,144 unique crops, 26,706 captions)	Relation understanding: true vs subject/object-swapped caption	ARO repo (subset of GQA / Visual Genome)
ARO VG-Attribution	28,748 cases (12,279 crops, 52,146 captions)	Attribute binding: true vs attribute-swapped caption	ARO repo
ARO COCO-Order	25,010 cases (5,000 images, 119,724 captions)	Word-order sensitivity: true caption vs 4 shuffled versions	Built from Karpathy test

NegCLIP training data. Each row holds an image path, one caption, 1–6 word-swapped negative captions (mean 1.5) and the indices of the image's 3 nearest-neighbour images. One negative caption and one neighbour image are drawn at random each epoch.

ARO tasks. VG-Relation and VG-Attribution crop the image to the box around the relation or attribute pair and offer two captions: "the horse is eating the grass" vs "the grass is eating the horse". Chance is 0.5. COCO-Order offers the true caption and four perturbations: nouns and adjectives shuffled, everything else shuffled, words shuffled within trigrams, and trigrams shuffled. Chance is 0.2. The ARO code redraws these perturbations at random on every load; we draw them once with seed 0 and cache them, so every model and every shift is scored on identical negatives.

Leakage check. NegCLIP was trained on Karpathy train plus restval. We verified by file name that Karpathy val and test share 0 images with the NegCLIP training file (Karpathy train shares 109,468). Karpathy val is NegCLIP's validation set, so it was used for checkpoint selection but never for a gradient step. We therefore fit Δ on val and report all results on test and ARO.

Download. Only the 10,000 Karpathy val and test images are needed for evaluation. Instead of the full 6.2 GB val2014 archive, we read the zip's central directory over HTTP range requests and fetched just those 10,000 files (about 1.6 GB). The 19 GB of full COCO 2014 images needed for fine-tuning were downloaded on the GPU server.

4. Methodology

Every model is encoded once per benchmark; all gap measurements, shifts and noise experiments then run on the cached, L2-normalised embeddings. This kept the CPU-only development machine usable and makes every shift experiment a matter of seconds.

4.1 Models

All seven models are OpenCLIP ViT-B/32 (512-d embeddings), so differences come from training only.

Model	Training	Hard neg. images	Hard neg. captions	Selected LR
clip	OpenAI CLIP, no fine-tuning	–	–	–
negclip	Released NegCLIP checkpoint	yes	yes	1e-6 (paper)

Model	Training	Hard neg. images	Hard neg. captions	Selected LR
negclip_gelu	Same weights, evaluated with standard GELU (as the ARO repo does)	yes	yes	1e-6
ft_plain	Ours: COCO fine-tuning only (the CLIP-FT control)	–	–	5e-6
ft_hardimg	Ours	yes	–	1e-6
ft_hardcap	Ours	–	yes	5e-6
ft_negclip	Ours: full NegCLIP reproduction	yes	yes	1e-6

The OpenAI weights were trained with QuickGELU, and NegCLIP was fine-tuned from them, so all models are built as `ViT-B-32-quickgelu`. The released NegCLIP file is a 1.8 GB training checkpoint with optimizer state; we extracted its 302 weight tensors with PyTorch's safe loader, allowing only the one numpy type it needed.

4.2 Fine-tuning the 2×2 control grid

NegCLIP differs from CLIP in three ways at once: COCO fine-tuning, hard negative images and hard negative captions. A single CLIP-FT control removes both negative types together. The 2×2 grid separates them, and ft_hardimg vs ft_negclip isolates hard negative captions at an identical image side of the batch.

We re-implemented the NegCLIP training fork (vinid/neg_clip) in one script, checked line by line against its data loader, loss, optimizer and scheduler. For a batch of N = 256 items, the batch is assembled as:

```
\text{images} = [I,\ I_{\text{hard}}], \qquad \text{texts} = [T,\
T(I_{\text{hard}}),\ T_{\text{neg}},\ T_{\text{neg}}(I_{\text{hard}})]
```

```
\mathcal{L} = \tfrac{1}{2}\Big[\mathrm{CE}\big(\text{images} \times \text{all
texts}\big) + \mathrm{CE}\big(\text{first } |\text{images}| \text{ texts}
\times \text{images}\big)\Big]
```

So negative captions only compete in the image→text direction. The resulting logit matrices are 256×256 (ft_plain), 512×512 (ft_hardimg), 256×512 (ft_hardcap) and 512×1024 (ft_negclip).

Recipe, identical for all four: 5 epochs; AdamW with betas (0.9, 0.98), eps 1e-6, weight decay 0.2 except on gains, biases and the logit scale; cosine LR schedule with 50 warmup steps; fp16 mixed precision; learnable logit scale clamped at 100; RandomResizedCrop (scale 0.9–1.0); seed 0. Each config was swept over LR ∈ {1e-5, 5e-6, 1e-6}, and the LR with the best final-epoch mean Recall@1 on the validation split was kept, as in the NegCLIP paper. After every epoch we also logged the validation gap, which shows how fine-tuning moves the gap.

4.3 Approach 01: measuring the gap

On Karpathy val and test, for each model we compute:

- **Gap size:** $\|\Delta\|$ from means and from coordinate-wise medians; cosine between the two centroids.
- **Cone width:** norm of each modality's centroid (close to 1 means a narrow cone).
- **Orthogonality:** the fraction of each modality's variance that lies along the gap direction $\hat{u} = \Delta/\|\Delta\|$. A random direction gets 1/512 ≈ 0.002; a value near that means the gap is nearly orthogonal to how embeddings vary (the Chowers condition).
- **Alignment and uniformity** (Wang and Isola): mean $\|x - y\|^2$ over matched pairs, and log $E \exp(-2\|u - v\|^2)$ per modality.
- **Similarity of matched and random pairs**, and **linear separability** (logistic regression accuracy at telling image from text embeddings).
- **PCA** of both modalities, as in Liang et al.'s Figure 1.

4.4 Approach 02: shifting the gap

Δ is fitted on Karpathy val and applied unchanged to every evaluation set. Embeddings are shifted along Δ and projected back onto the unit sphere:

$$\text{text mode: } y' = \frac{y + \alpha\Delta}{\lVert y + \alpha\Delta \rVert}, \qquad \text{both mode: } x' = \frac{x - \tfrac{\alpha}{2}\Delta}{\lVert \cdot \rVert},\ \ y' = \frac{y + \tfrac{\alpha}{2}\Delta}{\lVert \cdot \rVert}$$

$\alpha = 1$ closes the gap before re-normalisation; $\alpha < 0$ widens it. Two families of settings:

- **α sweep:** $\alpha \in \{-0.5, -0.25, 0, 0.25, 0.5, 0.75, 1.0\}$.
- **Target gap (shift by a set margin):** for each model, α is solved by bisection so that the remaining gap on Karpathy val equals 0.6, 0.4, 0.2 or 0.1. This compares models at equal remaining gap even when their original gaps differ.

Each is run with mean and median Δ, giving 44 settings per model.

4.5 Approach 04: evaluation

- **ARO:** accuracy that the true caption scores highest; macro accuracy over relations or attribute pairs (VG-Attribution: pairs with at least 25 cases), as in the ARO paper; and the margin between the true caption's score and the best wrong one.
- **Retrieval:** COCO 5k image→text and text→image Recall@1/5/10.
- **Robustness:** Gaussian noise with expected norm σ is added to both image and text embeddings after the shift, then re-normalised. $\sigma \in \{0, 0.25, 0.5, 1.0\}$, averaged over 3 seeds.

In total, 7 models × 44 settings × 4 benchmarks × 4 noise levels.

4.6 Implementation and compute

- **Encoding:** captions are about 15 tokens but padded to 77. CLIP's text encoder is causal and pools at the end-of-text token, so we run it only over the longest caption in each batch. This gives identical embeddings (max difference 3e-7) at 5× the speed on CPU.
- **CPU runs** (laptop, 6 cores, no GPU): CLIP and NegCLIP encoding, gap measurement and the shift sweep, about 5 h in total.
- **GPU runs** (gpu1.cse.iitb.ac.in, 3 × 48 GB GPUs): the 12 fine-tuning runs took 2.2 h (148 s per epoch for ft_plain up to 378 s for ft_negclip); encoding all 7 models took 9 min; the shift sweep for all 7 took 7 h on CPU cores.
- **Consistency:** clean scores for CLIP and NegCLIP from the GPU run match the CPU run to within 0.0002.

5. Results

5.1 Sanity checks: our pipeline reproduces published numbers

Every number we could check against the literature matches, except one NegCLIP figure that turned out to depend on an evaluation bug (Section 5.8).

Check	Ours	Published
CLIP gap $\|\Delta\|$, COCO	0.832	0.82 (Liang et al.)
CLIP COCO 5k i2t / t2i R@1	0.502 / 0.302	0.502 / 0.304 (Radford et al.)
CLIP VG-Rel / VG-Attr / COCO-Order	0.638 / 0.632 / 0.472	0.59–0.63 / 0.62 / 0.46

Check	Ours	Published
NegCLIP VG-Rel / VG-Attr / COCO-Order	0.838 / 0.721 / 0.825	0.81 / 0.71 / 0.86
Our ft_negclip vs released NegCLIP (VG-Rel / VG-Attr / COCO-Order)	0.840 / 0.725 / 0.838	0.838 / 0.721 / 0.825

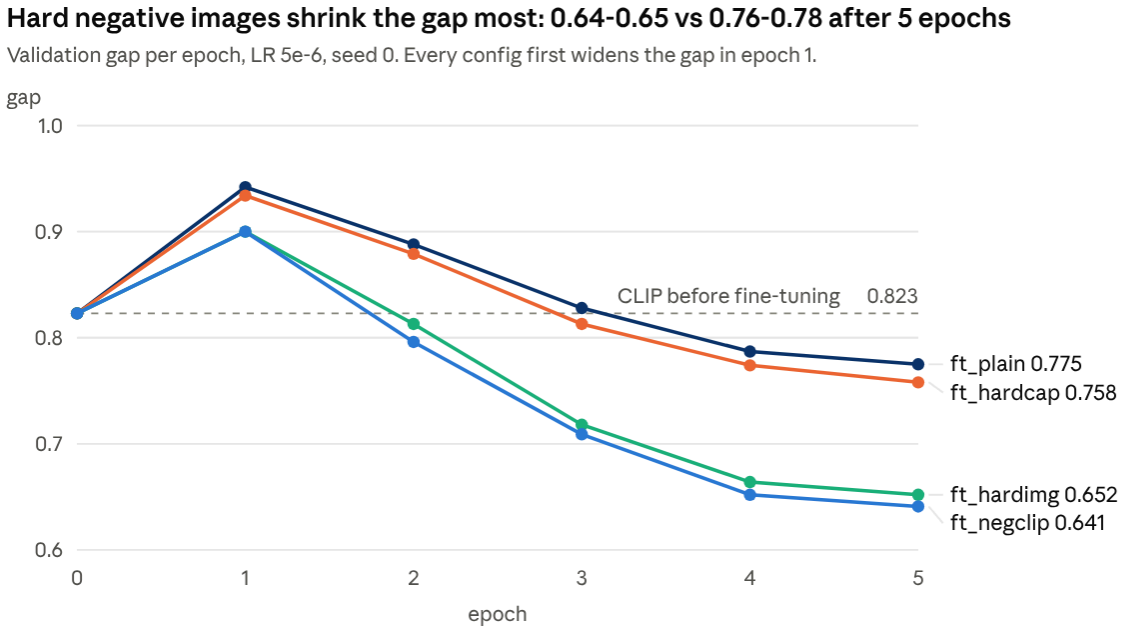
5.2 Hard-negative training does not shrink the gap

The released NegCLIP has a slightly larger gap than CLIP (0.855 vs 0.832 on Karpathy test). What changes is the text side: its text embeddings spread out and random image–text pairs are pushed much further apart. In every model the two modalities stay perfectly separable, and the gap direction carries only about 1.3–4× the variance of a random direction, so it is close to orthogonal to how embeddings vary.

Model	$\|\Delta\|$	Centroid cosine	Text centroid norm	Text uniformity	Matched cos	Random-pair cos
clip	0.832	0.304	0.723	−1.82	0.303	0.151
negclip	0.855	0.184	0.657	−2.16	0.292	0.083
ft_plain	0.780	0.127	0.556	−2.62	0.297	0.044
ft_hardimg	0.888	0.204	0.693	−1.98	0.297	0.101
ft_hardcap	0.765	0.096	0.550	−2.65	0.292	0.031
ft_negclip	0.883	0.192	0.687	−2.01	0.293	0.093

Karpathy test. Lower text centroid norm = wider text cone; more negative uniformity = more evenly spread. Linear separability of image vs text is 1.000 for every model. Note that ft_plain and ft_hardcap were selected at LR 5e-6 and the others at 1e-6, which matters for these numbers (Section 5.3).

5.3 During fine-tuning, the gap first widens, then shrinks with hard negative images



results/finetune_runs.csv (GPU run) · 4 configs × 6 validation checkpoints, LR 5e-6

Every configuration widens the gap in the first epoch (0.82 → 0.90–0.94), then shrinks it. The two configs with hard negative images end 0.11–0.13 lower than the two without; hard negative captions make a difference of only 0.01–0.02.

At the lowest LR (1e-6) all four configs end at almost the same gap (0.877–0.886). At LR 1e-5 the ordering is the same as at 5e-6: ft_plain 0.733, ft_hardcap 0.708, ft_hardimg 0.571, ft_negclip 0.555. Hard negative images also double the batch (512 vs 256 images), so this experiment cannot tell a harder batch from a bigger one.

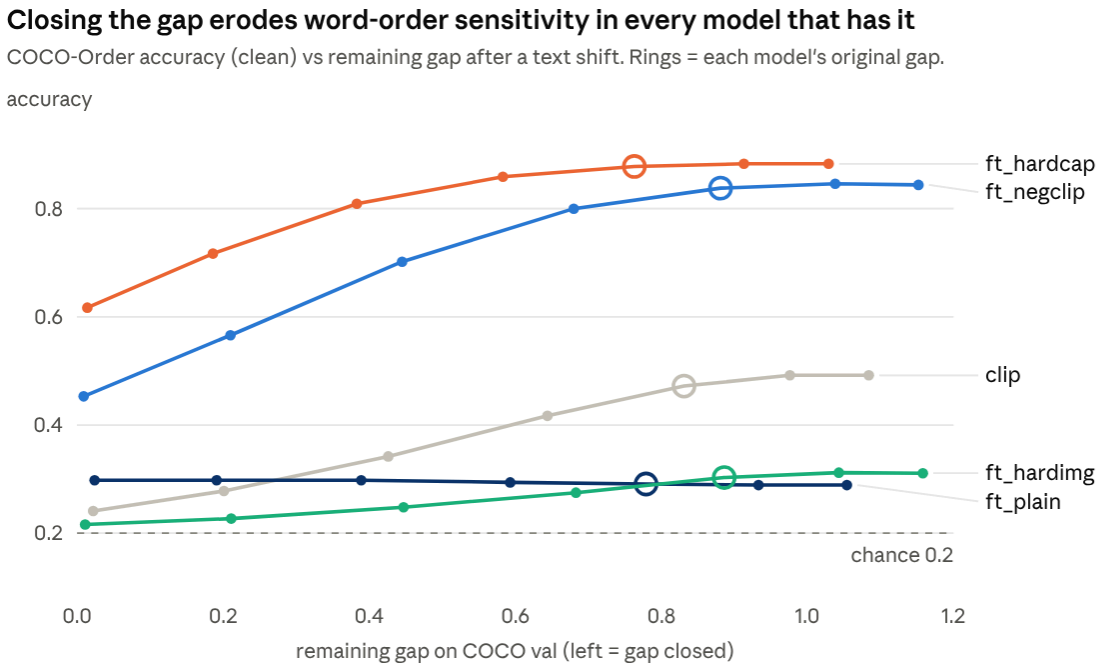
5.4 Compositionality comes from hard negative captions

Model	COCO-Order	VG-Attribution	VG-Relation	COCO i2t R@1	COCO t2i R@1
clip	0.472	0.632	0.638	0.502	0.302
ft_plain	0.291	0.647	0.542	0.618	0.454
ft_hardimg	0.303	0.650	0.554	0.607	0.451
ft_hardcap	**0.878**	**0.738**	**0.860**	0.599	0.454
ft_negclip	0.838	0.725	0.840	0.597	0.449
negclip (released)	0.825	0.721	0.838	0.593	0.448
negclip_gelu	0.868	0.705	0.811	0.561	0.412

Karpathy test and ARO, no shift. Macro accuracy for the VG tasks. Chance: COCO-Order 0.2, VG 0.5.

The two models with hard negative captions are the only ones that become compositional; hard negative images alone add almost nothing over plain fine-tuning. Plain COCO fine-tuning improves retrieval by 0.12–0.15 Recall@1 but makes the model less order-sensitive than CLIP (COCO-Order 0.47 → 0.29). Combined with Section 5.3, the size of the gap and compositionality are driven by different ingredients.

5.5 Closing the gap hurts order-sensitive models on clean data



results/shift_eval.csv (GPU run) · text shift, mean Δ, α from −0.5 to 1.0, no noise

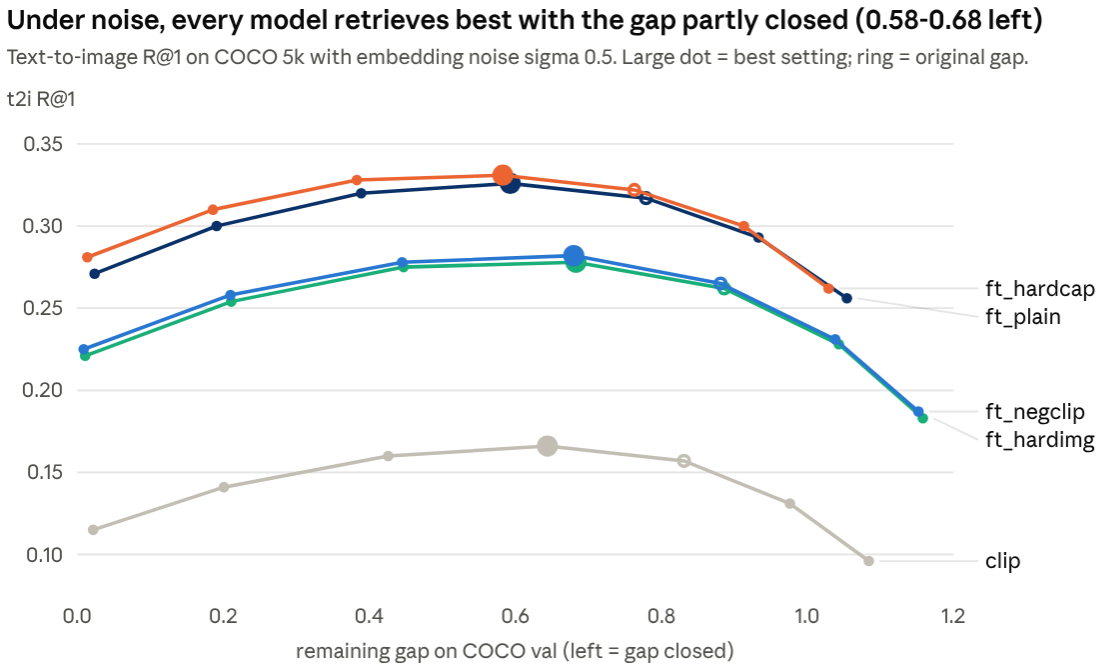
Fully closing the gap takes ft_negclip from 0.838 to 0.453 and CLIP from 0.472 to 0.241, close to chance. ft_plain is flat because it is already order-blind (0.29).

The effect of closing half the gap (α = 0 → 0.5, text shift) for every model:

Model	COCO-Order, clean	COCO-Order, σ = 0.5	VG-Relation, clean	VG-Relation, σ = 0.5	t2i R@1, clean	t2i R@1, σ = 0.5
clip	−0.131	−0.041	+0.020	+0.007	−0.044	+0.004
ft_plain	+0.007	+0.006	+0.057	+0.041	−0.029	+0.003
ft_hardimg	−0.055	−0.015	+0.032	+0.033	−0.042	+0.014
ft_hardcap	−0.070	−0.047	−0.006	−0.003	−0.024	+0.005
ft_negclip	−0.136	−0.048	−0.011	+0.014	−0.039	+0.013
negclip	−0.116	−0.043	+0.007	+0.002	−0.035	+0.011

Clean, the original gap is close to optimal for retrieval and COCO-Order. Closing it helps VG-Relation only for models trained without hard captions (+0.02 to +0.06). The symmetric "both" shift hurts more than the text-only shift at the same remaining gap (NegCLIP COCO-Order at gap 0.4: 0.627 vs 0.694), and mean vs median Δ differ by at most 0.01.

5.6 Under noise, a partly closed gap is best



results/shift_eval.csv (GPU run) · text shift, mean Δ, σ = 0.5, mean of 3 noise seeds

The gap between the two upper lines (ft_plain, ft_hardcap) and the two middle ones (ft_hardimg, ft_negclip) mostly reflects their different learning rates (5e-6 vs 1e-6), not the hard negatives (Section 6).

The same shape holds for image→text retrieval, COCO-Order and both VG tasks at σ = 0.25, 0.5 and 1.0. Widening the gap always makes noise robustness worse. The gain from partial closing is larger for the hard-negative models (+0.011 to +0.014 t2i R@1 at α = 0.5) than for CLIP and ft_plain (+0.003 to +0.005).

5.7 Why the shift changes which caption wins

Without re-normalisation, a text shift leaves image→text accuracy exactly unchanged (CLIP COCO-Order 0.472 → 0.472; NegCLIP 0.825 → 0.825), because the same constant $x \cdot \Delta$ is added to every candidate caption's score. With re-normalisation the shifted score is:

```
s'(x, y) = \frac{x^\top y + \alpha\, x^\top \Delta}{\lVert y + \alpha \Delta \rVert}
```

The numerator gains a large positive constant, and the denominator is smaller for captions that lie further on the text side of the gap direction, so those captions get boosted. Word-shuffled COCO-Order captions lie further on the text side than the true caption in 74% of cases for CLIP and 88% for NegCLIP. Closing the gap therefore systematically favours the shuffled negatives. VG-Relation's negatives are grammatical subject–object swaps whose projection matches the true caption (about 50/50), so VG-Relation barely moves. The gap direction partly encodes how natural a caption is, and hard-negative training strengthens that encoding.

5.8 An evaluation mismatch behind NegCLIP's reported COCO-Order score

NegCLIP was trained with QuickGELU activations: the training fork loads OpenAI weights through a builder that sets `quick_gelu=True`. The ARO repository, however, evaluates the checkpoint as a plain `ViT-B-32`, which uses standard GELU. With the activation it was trained with, NegCLIP scores 0.825 on COCO-Order; with the mismatched GELU it scores 0.868, matching the paper's 0.86. The mismatch costs 2–3 points on VG-Relation, VG-Attribution and retrieval. We report the QuickGELU model throughout.

6. Discussion and limitations

What the hypotheses came to.

- **H1 (hard-negative training changes the gap): only partly.** Hard negative captions reshape the text side (wider, more uniform text cone; random pairs pushed apart) but barely change the gap's size. The gap moves with hard negative images and with the learning rate.
- **H2 (closing the gap buys robustness for free): no.** The robustness gain is real for every model, and slightly larger for hard-negative models, but on clean data closing the gap costs retrieval and order sensitivity. The best gap under noise (about 0.4–0.6) is not the best gap clean (the original, about 0.8).
- **H3 (a uniform text shift barely changes caption rankings): wrong for order tasks.** The algebra holds without re-normalisation, but re-normalisation turns the shift into a penalty on captions that look "natural", which is exactly what COCO-Order tests.

Limitations.

1. **Learning-rate confound.** LR selection chose 5e-6 for ft_plain and ft_hardcap and 1e-6 for ft_hardimg and ft_negclip. The per-epoch validation logs compare all four at matched LR, but the benchmark and shift results compare checkpoints at different LRs. ft_plain and ft_hardcap look much more noise-robust in the figures, mostly because of their higher LR. Matched-LR checkpoints (ft_plain and ft_hardcap retrained at 1e-6) are needed before drawing per-config robustness conclusions.

2. **Batch-size confound.** Hard negative images double the number of images per batch (512 vs 256), so their effect on the gap mixes batch hardness with batch size.
3. **One seed.** Each fine-tuning run used seed 0; differences of a point or two between configs are within plausible seed noise.
4. **Definition of the orthogonal shift.** Our shifts move embeddings along the global gap vector and re-normalise. Chowers et al. may define their orthogonal shift differently; we have not yet compared against their exact definition.
5. **Benchmarks.** We did not yet evaluate hard positives (Kamath et al., 2024), SugarCrepe, or an out-of-domain retrieval set such as Flickr30k. NegCLIP and our controls were fine-tuned on COCO, so COCO retrieval is in-domain for them.
6. **Noise model.** Robustness is measured with isotropic Gaussian noise on embeddings, not with corruptions of real images or captions.

7. Conclusion and future work

Closing the modality gap does not help hard-negative-trained CLIP on clean compositional benchmarks: it costs NegCLIP-style models 0.07–0.14 COCO-Order accuracy for half the gap closed, and costs retrieval for every model. What it does buy is robustness: under embedding noise, a partly closed gap (0.4–0.6 remaining) beats the original gap for every model, and the gain is slightly larger for models trained with hard negatives.

The gap and compositionality are largely decoupled. Compositionality comes from hard negative captions; the gap's size moves with hard negative images (or the larger batch they bring) and with the learning rate, after an initial widening in the first epoch. The gap direction is not meaning-free either: it partly separates natural captions from word-shuffled ones, more strongly after hard-negative training, which is why a re-normalised shift damages order sensitivity.

Along the way we built a full, reproducible pipeline (data download, leakage check, cached encoding, gap metrics, shifts, evaluation, and a one-file NegCLIP reproduction that matches the released model), and found that the published NegCLIP COCO-Order score relies on evaluating the model with the wrong activation function.

Next steps, in order:

- [ ] Retrain ft_plain and ft_hardcap at LR 1e-6 (about 15–20 min each on gpu1) and re-run the evaluation, to remove the learning-rate confound.
- [ ] Add a control that keeps the batch at 512 images without hard negatives, to separate batch size from batch hardness.
- [ ] Add a re-normalisation-free shift and an exactly orthogonal shift, following Chowers et al.'s definition, as controls that leave image→text rankings fixed.

- [ ] Evaluate hard positives (Kamath et al., 2024), SugarCrepe and Flickr30k.
- [ ] Repeat the fine-tuning grid with 3 seeds for error bars.

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